Model	GSM8K	MATH-500	AIME2024	AIME2025	AMC2023	OlympiadBench	AVG
Llama-3.1-8B-Instruct	75.89	35.20	4.17	1.04	23.59	12.00	25.32
+ 20K QwQ-Distill-Data	82.41	60.80	4.38	8.33	32.97	25.48	35.73
+ 20K QwQ-Distill-Data + RL	91.51	69.80	8.96	4.17	37.97	25.93	39.72
+ 20K QwQ-Mole-Syn	84.31	50.20	5.21	1.67	32.34	20.00	32.29
+ 20K QwQ-Mole-Syn + RL	88.78	70.80	7.50	3.33	39.22	21.04	38.44
+ 35K QwQ-Mole-Syn + RL	90.30	68.40	10.00	4.38	39.84	24.15	39.51

Table 8 The reinforcement learning performance based on 3 initialization methods across 6 benchmarks.

the supervised pretraining/fine-tuning history).

F Details of Bond Shaping Function Analysis

This section provides a detailed description of the experimental settings for the geometric reshaping analyses presented in Fig. 13, corresponding to the three "bond" types in the Long CoT structure: Deep Reasoning (covalent bond), Self-Exploration (exploration bond), and Self-Reflection (stabilizing bond).

To analyze the function of different "bonds," we extracted step-level hidden representations from the penultimate Transformer layer of both Llama-3.1-8B-Instruct and our trained Mole-Syn model. For each reasoning step t, we computed its vector representation $\mathbf{h}_t$ via average pooling of its token embeddings. We then used t-SNE to visualize the semantic space and compare the representations $\mathbf{h}_t^{\text{orig}}$ and $\mathbf{h}_t^{\text{method}}$ for the same reasoning step across the two models.

F.1 Deep Reasoning: Densing the Primary Structure.

To quantify how Deep Reasoning densifies the "primary structure," we use the same development-set split as in the main text. We compare two training modes: (1) *baseline*, which uses standard short CoT prompting by standard Llama-3.1-8B-Instruct; and (2) *deep-reasoning-trained*, which uses the Long CoT paradigm with the Deep Reasoning transfer probability set to 50%, while the remaining probability mass follows the other behaviors in our default setting. For each input sample, we generate 10K training instances and, under each mode, sample 100 reasoning trajectories from OlymBench. From these trajectories, we extract the step-level baseline representations $\{h_1, \ldots, h_T\}$ and deep-reasoning-trained representations $\{\tilde{h}_1, \ldots, \tilde{h}_T\}$.

To characterize the compactness of the "primary To characterize the compactness of the "primary structure" in semantic space, we compute the Minimum Enclosing Ball (MEB) over all step representations from the sampled trajectories:

$$S = \{\tilde{h}_1, \ldots, \tilde{h}_{T_N}\}, \tag{49}$$

where T_N denotes the total number of extracted steps aggregated across N trajectories for the given sample set. We compute the MEB in a shared three-dimensional embedding space obtained by t-SNE, which we use consistently across modes for visualization and for a comparable notion of geometric spread. The procedure is as follows. First, we reduce each step representation from the original high-dimensional embedding space to $d = 3$ using t-SNE. Second, we apply an approximate MEB solver based on Welzl's algorithm to obtain the center $c \in \mathbb{R}^3$ and radius r. Third, we define the corresponding volume in $d = 3$ Euclidean space as:

$$V = C_d \cdot r^d, \tag{50}$$

where C_d is the unit-ball volume constant (a fixed scaling factor that cancels in relative comparisons). We then compute the arithmetic mean of V over all development-set samples, denoted as V_{base} (Baseline) and V_{deep} (Deep Reasoning). The volume reduction reported in Fig. 13(a) is:

$$\Delta_{\text{Deep}} = \frac{V_{\text{base}} - V_{\text{deep}}}{V_{\text{base}}} \times 100\%. \tag{51}$$

A positive Δ_{Deep} indicates that Deep Reasoning contracts the semantic volume of the core logical backbone while preserving comparable semantic coverage of the task.

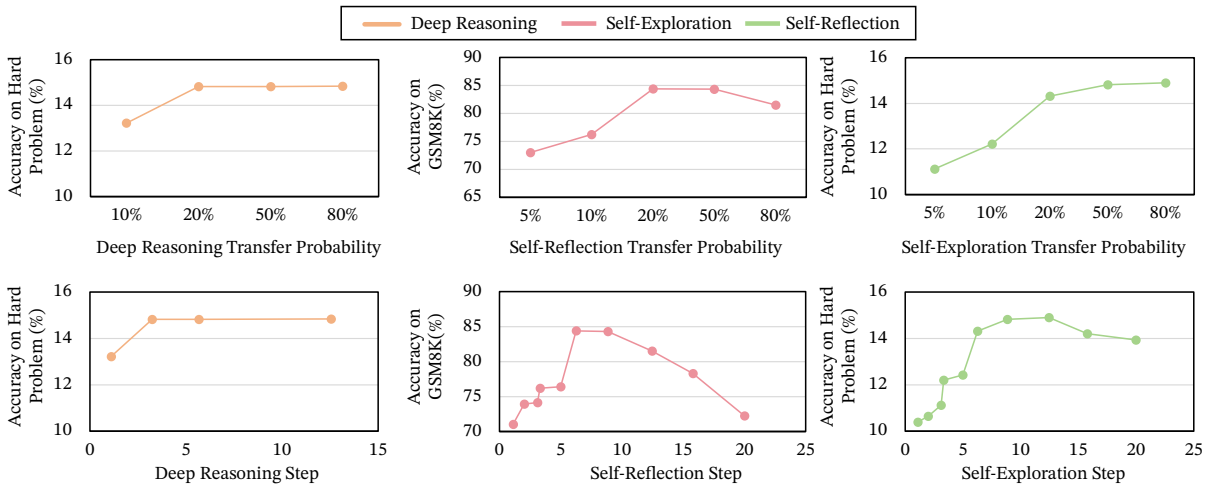

Figure 16 Comparative analysis of performance under different reasoning bond ratios.

F.2 Self-Exploration: Expanding the Logical Space.

We use the same development-set split and the same two training modes as above: (1) *baseline* (short CoT or direct answering, without Long CoT) and (2) *self-exploration-trained* (Long CoT with self-exploration transfer probability >50%, with other behaviors unchanged). For each input sample, we generate 10K training instances and sample 100 reasoning trajectories from OlymBench under each mode, then extract the corresponding step representations. Let V_{exp} denote the mean MEB volume computed from the step representations used for the *self-exploration-trained* model. We quantify the relative volume change as:

$$\Delta_{\text{Exp}} = \frac{V_{\text{exp}} - V_{\text{base}}}{V_{\text{base}}} \times 100\%. \tag{52}$$

Then, a positive Δ_{Exp} corresponds to an expansion of the explored logical space relative to the baseline.

F.3 Self-Reflection: Densifying and Stabilizing the Logical Results.

We follow the same data construction and representation extraction protocol as above. To measure the contraction induced by Self-Reflection, we compute MEB volumes before and after the reflection step, denoted by V_{pre} and V_{post}, respectively. We then report the relative contraction as

$$\Delta_{\text{Reflect}} = \frac{V_{\text{pre}} - V_{\text{post}}}{V_{\text{pre}}} \times 100\%, \tag{53}$$

where a larger value indicates a stronger "folding" effect in semantic space after reflection.

To further examine the "suppression of inconsistent branches," we conduct an auxiliary clustering analysis (not shown in the main text). Specifically, we cluster the pre- and post-reflection point sets, S_{pre} and S_{post}, using an α-threshold grouping method. We observe that, after reflection, intra-cluster distances decrease substantially, while inter-cluster distances remain stable or increase slightly. This pattern suggests that the dominant semantic cluster becomes more compact, whereas inconsistent branches are pruned or weakened. Consistent with these findings, the overall semantic volume decreases from 35.2 to 31.2, indicating that Self-Reflection guides the structure toward a more stable solution manifold.

G Analysis of the impact of length and diversity in reasoning behaviors

To evaluate whether comparable performance gains can be achieved merely by regulating the length and frequency of Self-Reflection steps, we conducted a detailed analysis illustrated in Fig. 13, covering the three distinct "bond" types that constitute the Long CoT framework.

G.1 Impact of bond-enhanced behaviors on performance

We further examined how different reasoning behaviors influence overall model performance (Fig. 16). For complex queries, improvements are primarily driven by reasoning depth rather than the sheer number of steps, once a minimum threshold is achieved. A higher proportion of reflective behaviors tends to enhance performance on challenging tasks but can introduce "overthinking," which reduces accuracy on simpler ones. Similarly, excessive exploratory behavior on complex tasks often impedes convergence—indeed, 62.7% of AIME cases exhibited extended reasoning without reaching a clear conclusion. In contrast, simple queries generally benefit from more direct exploration, leading to greater accuracy. Across all models, performance consistently peaks at a balanced distribution of behaviors, suggesting the existence of a stable and task-invariant optimal configuration.

G.2 Effect of bond length on reasoning quality

We also investigated how varying bond lengths affects performance while maintaining constant behavioral ratios. As shown in Fig. 16, increasing both the number of steps and the length of individual reasoning bonds generally strengthens performance, particularly in deep reasoning scenarios. However, overly long exploration bonds can induce semantic drift and reduce the precision of final outputs. The optimal bond length appears to depend on task difficulty: shorter bonds tend to perform better on simple problems, whereas longer ones are more advantageous for tackling complex challenges.

H Details of LLM Structure Reconstruction

H.1 Reasoning compression or summarization data collection.

For the analysis about broken structure reconstruction, we treat Gemini-2.5-Pro-Thinking and Claude-4-Sonnet as black-box teacher models that can emit long chain-of-thought (CoT) traces. We query both APIs with identical user prompts and task instructions, and enable their "thinking" content, which are much shorter than other reasoning LLMs with Long CoT.

H.2 Summarize Reasoning Process Analysis.

To emulate the protection strategies used in private LLMs, we summarize the full Long CoT traces from QwQ-32B and OpenAI-OSS-120B into concise summaries by Qwen2.5-32B. This procedure removes the long-range reasoning structure of the original CoT, thereby reducing the amount of recoverable step-by-step rationale. The summarization prompt is as follows:

Prompt for Summarization

You are an expert summarizer. Below is a Long Chain-of-Thought reasoning trace generated by an AI model to solve a complex problem. Your task is to compress this reasoning process into a concise summary.

Input Long Chain-of-Thought Trace:
[Insert Full Trace Here]
Summary:

We trained the model on the summarized 20K samples. The corresponding results are reported in Table 4. All other analytical experiments followed the same configuration as in the preceding sections, except for the model and the reported outputs.